ASCII MASTER FLOW — PANEL 1/12: Agricultural geography as a natural-human-market system

```
+-------------- AGRICULTURAL GEOGRAPHY ----------------+
| Where does production occur, why there, how organised,|
| and how does the pattern change through time?         |
+--------------------------------------------------------+
                         |
       +-----------------+------------------+
       |                 |                  |
NATURAL BASE         HUMAN SYSTEM       MARKET SYSTEM
climate              labour/holding     demand
relief               technology         transport
soil                 tenure/capital     storage/processing
water                institutions       price and trade links
       \                 |                  /
        +---------- FARMING SYSTEM ----------+
                         |
 location -> seasonality -> intensity -> surplus -> value chain
                         |
 spatial gains and costs feed back to soil, water, labour and region

Six controls: climate, relief, soil, water, labour, market/transport.
MUST REMEMBER: Agricultural geography reconstructs land use through physical controls, tenure,
  labour, technology, markets and policy; it compares subsistence/commercial,
  intensive/extensive, plantation, mixed, dairy, Mediterranean, shifting, pastoral and von
  Thünen location logics before mapping India's crop seasons and agro-regions.
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```

ASCII MASTER FLOW — PANEL 2/12: Farming-system classification by ecology, labour and market

SYSTEM	ORGANISATION	TYPICAL SETTING
Shifting cultivation	clearing + fallow	humid forest margins
Wet-rice intensive	small, labour, water	monsoon valleys
Nomadic pastoralism	mobile herds	arid sparse pasture
Mixed farming	crop-livestock link	temperate regions
Mediterranean	winter rain system	west-coast margins
Plantation	estate + export crop	tropical highlands
Commercial grain	mechanised vast farms	temperate plains
Dairy/market garden	perishable + urban	city hinterlands

CLASSIFICATION AXES

subsistence/commercial | intensive/extensive | crop/livestock mix

sedentary/mobile | small/large scale | local/export orientation

High output per hectare is not the same as commercial orientation.

ASCII MASTER FLOW — PANEL 3/12: Von Thunen rings and the survival of distance

```
                      CENTRAL MARKET
                 +------------------------+
                 | dairy / vegetables     | high decay, frequent trips
                 +------------------------+
                 | forestry / fuelwood    | heavy transport burden
                 +------------------------+
                 | field crops            | lower perishability
                 +------------------------+
                 | livestock / ranching   | extensive outer use
                 +------------------------+

distance up -> freight/reliability cost up -> bid rent generally down
perishability + delivery frequency + land rent decide near-market use
                         |
MODERN MODIFIERS
cold chain + expressway + processing + digital markets + policy
                         |
stretch, distort or split rings; they do not abolish cost and access

Answer route: assumptions -> mechanism -> modification -> surviving use
              -> limits from relief, multiple markets and institutions.
```

ASCII MASTER FLOW — PANEL 4/12: World crop regions and ecological requirements

CROP	ECOLOGICAL BUNDLE	BROAD GEOGRAPHY
Rice	warmth + managed water	monsoon valleys/deltas
Wheat	cool growth, dry ripening	temperate and winter belts
Cotton	frost-free warmth	subtropical plains/plateaus
Jute	humid alluvium + retting	eastern deltaic belt
Tea	humid acidic slopes	Asian tropical highlands
Coffee	shaded drained uplands	tropical uplands
Millets	variable lower rainfall	semi-arid drylands

REGIONAL LOGIC

Monsoon Asia -> intensive rice | temperate plains -> commercial grain
Mediterranean margins -> winter crops, vines, olives and fruits
Tropical estates -> plantation crops | city belts -> dairy and vegetables

A crop-soil link is an association, not an exclusive rule.

ASCII MASTER FLOW — PANEL 5/12: India's agro-climatic mosaic, Green Revolution and food chain

```
INDIA'S AGRICULTURAL MOSAIC
North-west       : irrigated wheat-rice and procurement core
Eastern deltas   : rice, jute and water-rich farming
Dryland Deccan   : millets, pulses, oilseeds and variable rainfall
Western plateau  : cotton and mixed irrigated-dryland systems
Highlands        : tea, coffee, spices and horticulture
Mountains        : horticulture, pastoral and terrace systems
                         |
GREEN REVOLUTION PACKAGE
HYV + assured irrigation + fertiliser + credit + extension + procurement
                         |
early concentration: Punjab + Haryana + western Uttar Pradesh
                         |
higher grain output -> procurement -> central pool -> NFSA/PDS movement
                         |
SECOND-GENERATION FEEDBACK
water-table decline + nutrient imbalance + residue pressure
+ rice-wheat lock-in + regional divergence

Productive success and ecological stress can coexist in the same region.
```

ASCII MASTER FLOW — PANEL 6/12: Fishing, forestry, mining and quarrying through physiography

ACTIVITY	PHYSICAL BASE	HUMAN FILTER
Fishing	shelf, nutrients, mixing	harbour, cold chain
Inland fish	river, tank, reservoir	rights, seed, markets
Forestry	climate, altitude, species	access, tenure, rules
Mining	ore body or fuel basin	grade, power, transport
Quarrying	accessible bulk material	demand, regulation

```
PHYSIOGRAPHIC SEQUENCE
resource occurrence -> accessibility -> extraction/harvest -> processing link
                         |
technology + transport + institutions + rights determine realised activity

Indian cues:
central/NE forests -> timber and NTFP | shields -> many metallic ores
sedimentary basins -> coal/limestone | margins -> offshore hydrocarbons

Primary extraction ends before manufacturing begins.
```

ASCII MASTER FLOW — PANEL 7/12: Sustainable agriculture and intervention stack

```
REGIONAL PRODUCTION PRESSURE
monsoon variability + fragmented holdings + unequal irrigation
+ soil decline + water stress + market risk
                         |
RESPONSE STACK
soil and water conservation
-> crop-water alignment and diversification
-> efficient irrigation and resilient seed systems
-> storage, aggregation, processing and cold chain
-> credit, insurance, extension and market access
-> procurement reform with food-security safeguards
-> local monitoring of income, nutrition and ecology

EVIDENCE FIREWALL
estimate is not final | output is not procurement | procurement is not stock
stock is not consumption | MSP announcement is not actual purchase

SCHEME STATUS
announcement -> approval -> guidelines -> selection -> funds
-> physical output -> service outcome -> long-term impact

Technology raises output only where water, knowledge and institutions converge.
```

ASCII MASTER FLOW — PANEL 8/12: Map-answer and Mains spine for agricultural geography

```
QUESTION: Why is agricultural change spatially unequal?
                             |
1. DEFINE: location, seasonality, organisation and change
                             |
2. MAP: irrigated NW | eastern deltas | dry Deccan | plantation hills
                             |
3. BUILD CONTROL MATRIX
   climate + relief + soil + water + labour + market/transport
                             |
4. EXPLAIN SYSTEM
   subsistence -> surplus -> specialisation -> processing -> value chain
                             |
5. ADD NAMED EVIDENCE
   Green Revolution core, procurement geography or physiographic activity
                             |
6. EVALUATE
   productivity + food security versus water, soil and regional inequality
                             |
7. QUALIFY AND CONCLUDE
   diversify only with purchase assurance, processing and transition support

Claim -> evidence -> spatial mechanism -> distributional/ecological limit
      -> region-sensitive verdict.
```

ASCII MASTER FLOW — PANEL 9/12: Evidence ladder and confidence control

EVIDENCE LADDER

1. = grounded source = synthesis / exam heuristic = verified dated...
2. Source/date: PIB, 28 May 2025 -Cabinet approval of MSP for Kharif...
3. central Indian and north-eastern forest belts and is a principal...

VERDICT: Source type controls the strength and limits of the historical claim.

ASCII MASTER FLOW — PANEL 10/12: Examiner traps and contested boundaries

CLOSE DISTINCTIONS

1. Do not turn a dated official estimate into a timeless fact.
2. Do not treat a scheme dashboard as an impact evaluation.
3. Do not begin with India-specific complexity before the core location...

VERDICT: Qualification earns marks; false certainty is a hard failure.

CLOSE DISTINCTION: Kharif, rabi and zaid are seasons rather than crop essences; net sown area differs from gross cropped area, cropping intensity from yield, MSP announcement from procurement, food security from cereal self-sufficiency, and irrigation potential from realised equitable water delivery.

CLOSE DISTINCTION: Kharif, rabi and zaid are seasons rather than crop essences; net sown area differs from gross cropped area, cropping intensity from yield, MSP announcement from procurement, food security from cereal self-sufficiency, and irrigation potential from realised equitable water delivery.

ASCII MASTER FLOW — PANEL 11/12: Integrated answer spine and qualified conclusion

```
ANSWER SPINE
1. Practice contract Every solved item has demand decoding, a detailed...
2. NATURAL BASE -> FARMING SYSTEM -> LOCATION -> MARKET LINK -> CHANGE...
3. In Prelims, UPSC separates close classifications. In Mains, it...
VERDICT: Claim, named evidence, analysis and qualification lead to a graded...
EVIDENCE LIMIT: Punjab–Haryana wheat-rice, western Maharashtra sugarcane, Malwa
  cotton/soybean, Assam tea, Kerala rubber/spices, Karnataka coffee, eastern rice, dryland
  Deccan millets/pulses and horticultural belts must be explained through water, soil, labour,
  market and policy interactions, with Agriculture Census/land-use/production data
  source-dated.
```


```
EVIDENCE LIMIT: Punjab–Haryana wheat-rice, western Maharashtra sugarcane, Malwa
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  market and policy interactions, with Agriculture Census/land-use/production data
  source-dated.
```

ASCII MASTER FLOW — PANEL 12/12: Evidence ladder and confidence control — DEEP-REVIEW SYNTHESIS 2

EVIDENCE LADDER

1. = grounded source = synthesis / exam heuristic = verified dated...
2. Source/date: PIB, 28 May 2025 -Cabinet approval of MSP for Kharif...
3. central Indian and north-eastern forest belts and is a principal...

VERDICT: Source type controls the strength and limits of the historical claim.